find names and job titles of people mentioned on a Web site. For example, such programs can find "Jane Smith, Vice President of Marketing," or "Joe Jones, CEO," according to Mitchell. He even goes so far as to say that you can go to a search engine and type in, "Show me a list of universities that offer meteorology as a major and order them by student-to-faculty ratio"!

Is that an exaggeration? Coming from a former president of the AAAI, we're tempted to think not. However, AI claims are traditionally exaggerated, and there have been brilliant people in the past who predicted that a computer would become as intelligent as an average human being by 1980. AI claims, therefore, sometimes need to be taken with a pinch of salt.

An example of a current search engine that has AI claims to fame is Accoona. From the "Artificial Intelligence" link on Accoona: "Accoona Artificial Intelligence is a Search Technology that understands the meaning of search queries beyond the conventional method of matching keywords. This user-friendly technology, merging online and offline information, delivers more relevant results and enhances the user experience...

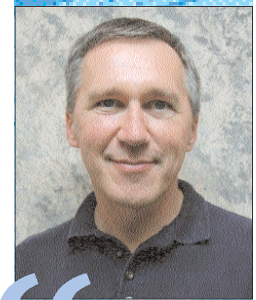
"Accoona's AI uses the meaning of words to get you better searches. For example, when you type five keywords in a traditional search engine, you're going to get every page that has all five keywords, no more, no less. With Accoona's AI Software, which understands the meaning of the query, the user will get many additional results. Accoona's AI also super-targets your search. For example, within a query of five keywords, Accoona AI allows the user to highlight one keyword, and will rank the search results starting by every page where the meaning of that one keyword is more important than the meaning of the other four keywords."

So does Accoona live up to the hype? We typed in "When was Britney born?", again, into Accoona, and first appeared sponsored results about Born shoes. The top results all brought up the song *Born to make you happy*. No AI at all here. Accoona doesn't parse what we type into it, and it doesn't recognise our query as a question—leave alone supplying the answer.

What's throwing so many engines astray is the song *Born to make you happy*, which almost always shows up before links containing the birthdate. Another reason is that common words are ignored, so that means 'when' and 'was' are ingored—precluding the possibility of parsing the query. This illustrates what plagues search engines today: they search mostly on keywords, with little attention to what your query means or what a page is trying to say.

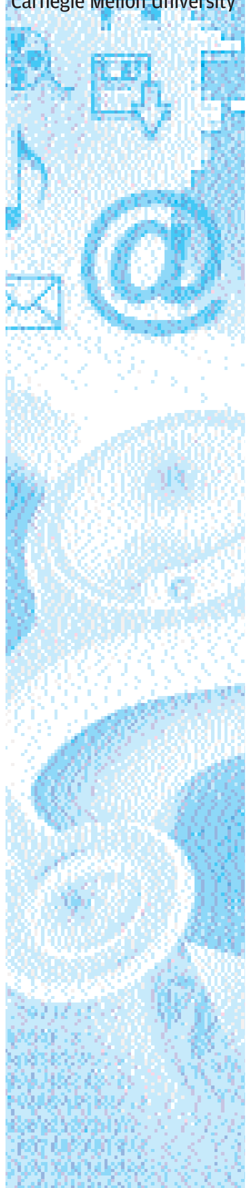
This is changing; Google already uses LSI, and is giving it more and more importance (see section '*Latent Semantic Indexing*' for more).

Reviews of Accoona are mixed, but the general consensus is that the database is not large enough yet for it to replace any of the biggies. And in our experience, the artificial intelligence doesn't show up. The time isn't ripe yet—and perhaps we'll see better AI implementations in the years to come.



"We can already develop computer software that can examine a Web page, and find names of people, dates and locations

Tom Mitchell
Director,
Center for Automated Learning and Discovery,
School of Computer Science,
Carnegie Mellon University



Personalisation

Some argue that the future of search engines lies in personalisation. And that means giving up some of our personal data, such as browsing habits, to the search engines. People paranoid about sharing such information might not like the idea; they're the same people who wouldn't want to use GMail because GMail 'reads' your mails. But personalisation could just be the way to go.

Google recently launched the personalised version of its search, at **http://labs.google.com/personalized**. You create a profile of your interests, and Google returns your results based on those, and more importantly, Google remembers you over sessions if you log in with your Google account. It is thus able to offer you better and better results over time, because, simply speaking, each time you conduct a search and click on a link, you're giving it a better idea of who you are and what it is that you're likely to want displayed as search results. This is rudimentary AI at work. For example, if you've often conducted searches on home improvement and on glass, a personalised search engine is more likely to interpret your "windows" keyword as meaning glass windows, rather than the operating system.

Craig Silverstein, a data mining researcher and now a director at Google, has a quote about personalised search: "It's clear that a list of links, though very useful, doesn't match the way people give information to each other... How can the computer become more like your friend when answering your questions? That means giving direct answers to questions, extracting data from online sources rather than giving links to Web pages. It also means doing a better job of divining what the searcher is looking for, tailoring results more closely to what, based on past experience, appear to be the user's particular interests."

Clustering

There are some in the know who argue that clustering, not personalisation, will be the future of search. A search engine called Clusty currently clusters search results admirably—a search on "Britney Spears" brought forth the following clusters: "Britney Spears pictures (53 sites), Nude (28 sites), Artists, Art (24 sites), Fan (17 sites), MP3 (10 sites), and so on. And these clusters actually had relevant documents in them! The 'Pictures' cluster linked to pictures sites, the 'Fan' cluster led to fan sites, and so on.

Of course, you might ask, "Why not supply all the keywords—such as 'Britney Spears Fan Sites'—in the first place?" The answer is that the clusters bring out related things of interest that you might miss out with your keywords.

In other words, *you* might not know exactly what *you're* looking for. In this example, the 'Federline' cluster contained 10 documents—Kevin Federline is Britney's husband, which we didn't know until we'd conducted this search on Clusty. And naturally, someone interested in Britney is likely to be interested in the related topic of her husband as well.

Seo-scoop.com has this to say about clustering: "Clustering related search terms into groups